

PAYMENT TRANSACTION ANOMALY DETECTION

Using Machine Learning

Technical Project Report

Project Domain	FinTech / Machine Learning
ML Task	Binary Classification
Application	Streamlit Web Prototype
Dataset	Synthetic educational transaction dataset
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An end-to-end educational machine learning prototype for identifying unusual payment transaction patterns.

Important: this is an educational prototype using synthetic data. It is not a production financial fraud-detection system.

1. Abstract

Payment systems contain a large number of routine transactions, while a smaller number may behave differently from the usual pattern. This project explores how machine learning can classify payment transactions as normal or anomalous using transaction and behavioural information.

The work follows a complete workflow: dataset inspection, data preparation, exploratory analysis, feature engineering, model training, evaluation, error analysis and deployment through a lightweight Streamlit application. Logistic Regression, Decision Tree and Random Forest were considered as classification models.

The dataset contains 10,000 synthetically generated educational transaction records. The purpose is to demonstrate a reproducible machine-learning workflow rather than claim real-world financial fraud performance.

2. Introduction

A transaction may be unusual when its characteristics differ from patterns normally observed in the data. Examples include an unusually large amount, a new device, a change of country, or an unusually high transaction frequency.

The aim is not to make a final decision about whether a real payment is fraudulent. Instead, the project demonstrates how a labelled dataset can be used to build a classifier that provides a first-level prediction for a new observation.

3. Problem Definition

This is a binary classification problem. The target variable is **is_anomaly**, where 0 represents a normal transaction and 1 represents an anomalous transaction.

A major challenge is class imbalance. Most observations are normal, while anomalous observations are less common. Therefore, accuracy alone is not sufficient; precision, recall, F1-score and the confusion matrix are important for understanding false alarms and missed anomalies.

4. Objectives

The project objectives are:

- Inspect and understand the transaction dataset.
 - Check missing values and duplicate records.
 - Study class balance and transaction patterns.
 - Create meaningful behavioural and time-based features.
- Train and evaluate suitable classification models.
- Analyse false positives and false negatives.
- Build a simple Streamlit interface for new transaction predictions.

5. Dataset Description

The project uses a synthetic educational dataset created specifically for this prototype. It contains 10,000 records and 16 original columns. The target variable is **is_anomaly**.

Dataset distribution: **9,490 normal** transactions and **510 anomalous** transactions. This corresponds to 94.9% normal and 5.1% anomalous observations.

The dataset is synthetic and is not being presented as a Kaggle, bank or customer dataset. This distinction is kept explicit for transparency and reproducibility.

6. Main Features

The model uses transaction amount, merchant category, country, home country, channel, time since previous transaction, transaction frequency, average amount over 30 days, account age, distance from home and new-device information.

Additional features were engineered: **amount_ratio**, **country_changed**, **high_frequency**, **transaction_hour**, **day_of_week** and **transaction_month**.

7. Methodology

The modelling workflow was: load and inspect data → check quality → perform EDA → engineer features → separate X and y → use a stratified 80:20 train-test split → preprocess numerical and categorical features → train classifiers → evaluate predictions → save the trained model → connect it to Streamlit.

Numerical features were standardized and categorical variables were one-hot encoded. The preprocessing was kept inside the machine-learning pipeline so that the application can apply the same transformations during prediction.

8. Exploratory Data Analysis

The dataset was checked for missing values and duplicate records. The project found no missing values and no duplicate rows in the working dataset.

The notebook examined transaction amount, transaction frequency, distance from home, new-device usage, merchant categories and numerical relationships. The target distribution clearly showed that anomalous transactions form a minority class.

9. Machine Learning Models

Three classification approaches were developed: **Logistic Regression** as a simple baseline, **Decision Tree** to capture non-linear rules, and **Random Forest** as an ensemble of decision trees.

The models were evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC where appropriate, and Confusion Matrix.

10. Evaluation and Error Analysis

The recorded Logistic Regression confusion matrix on the test set was:

Actual Normal → Predicted Normal: **1214**

Actual Normal → Predicted Anomalous: **684**

Actual Anomalous → Predicted Normal: **55**

Actual Anomalous → Predicted Anomalous: **47**

These values correspond to approximately **63.05% accuracy**, **6.43% precision**, **46.08% recall** and **11.28% F1-score**. The relatively high number of false positives demonstrates why accuracy by itself can give an incomplete picture for an imbalanced dataset.

The notebook also evaluates the Decision Tree and Random Forest models. Their exact metric values should be taken from the final notebook output rather than reconstructed from unclear screenshots.

11. Streamlit Application

A working Streamlit prototype was created as the user-facing part of the project. It accepts transaction characteristics such as amount, merchant information, account history, distance, device status and time-related values.

After the user clicks the prediction button, the application displays a clear result such as **Normal Transaction** or **Anomalous Transaction**, together with an anomaly probability. The application was tested locally in a web browser and produced a prediction result.

12. Project Files

Payment_Transaction_Anomaly_Classification.ipynb — main notebook.

payment_transaction_anomaly_dataset.csv — dataset.

payment_anomaly_model.pkl — trained model/pipeline.

app.py — Streamlit application.

requirements.txt — required Python libraries.

README.md — project documentation.

13. Discussion

One of the main lessons from this project is that anomaly classification is not simply about obtaining a high accuracy value. When the anomalous class is small, a model can appear accurate while still missing anomalous observations or generating many false alarms.

The confusion matrix and class-specific metrics therefore provide useful information about how the model behaves. The Streamlit application then turns the trained pipeline into an easy-to-demonstrate prediction tool.

14. Limitations

- The dataset is synthetic and does not represent real banking transactions.
- The project is an educational prototype, not a production fraud-detection system.
- Real financial applications would require stronger validation, security, monitoring and governance.
- Model performance on this dataset should not be interpreted as real-world financial risk performance.

15. Future Scope

Possible improvements include threshold optimization, additional anomaly-detection techniques, larger validated datasets, explainable AI methods, real-time monitoring and periodic model retraining.

16. Conclusion

This project demonstrates an end-to-end machine-learning workflow for payment transaction anomaly classification. It combines data analysis, statistical thinking, feature engineering, machine learning and application development in one reproducible prototype.

The project also shows why imbalanced classification should be evaluated using more than accuracy. Precision, recall, F1-score and confusion-matrix analysis help explain the behaviour of the classifier in a

more meaningful way.

17. Reproducibility and References

The implementation uses free/open-source Python tools including Pandas, NumPy, Matplotlib, scikit-learn, Joblib and Streamlit.

Project specification: Learn Depth Academy LLP, Track 1 – Final Capstone, Problem 26: Payment Transaction Anomaly Classification. The specification calls for source code, notebook, dataset/source citation, trained model, requirements file, README, Streamlit application, technical paper and presentation, and requires submitted results to be reproducible and explainable by the student.

Declaration: This report documents the educational workflow and synthetic dataset used in the project. No real bank/customer transaction data is claimed.